

Project Title	Development of an FM Radio Receiver System		
Course Code	ECE213	Course Name	Electrical circuits II
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Problem Summary	This project develops a complete FM radio receiver system capable of picking up FM broadcast signals, tuning to a desired station, and delivering audio through a speaker. The system integrates four subsystems into one working prototype: • A wire antenna (75 cm) to capture FM signals across the 88–108 MHz broadcast band. • An FM tunable receiver using the TDA7000 IC, which integrates RF amplification, mixing, IF filtering, and FM demodulation, with an external LC tank circuit for station selection. • A Class AB audio amplifier implemented with the LM386N IC, whose internal output stage is a complementary push-pull Class AB design, delivering amplified audio to an 8Ω speaker. • A bonus regulated AC-to-DC power supply (transformer → bridge rectifier → 7809 regulator → 9V DC) replacing the battery with a stable mains-powered source. Key challenges include maintaining low-noise signal reception at FM frequencies, achieving stable LC tuning across the full FM band, and ensuring clean audio amplification without crossover distortion. The system was built and tested on a breadboard, demonstrating the complete signal chain from electromagnetic wave reception through demodulation to audio output.		
Methodology	The project followed a structured four-phase approach: 1. Component Selection: The TDA7000 FM receiver IC was selected for its high integration — RF amplifier, local oscillator, mixer, IF amplifier, limiter, FM demodulator, and mute logic in a single 18-pin DIP package. The LM386N audio amplifier was selected for its built-in Class AB push-pull output stage and low-voltage operation (4.5–12V). 2. LC Tuning Circuit Design: An LC tank circuit was designed using a 3.5-turn air-core coil (~40.5 nH) and a 0–60 pF variable and covering the full 88–108 MHz FM band. 3. Simulation (LTspice): Circuit stages were simulated in LTspice to verify the LC resonant frequency at ~100 MHz and to confirm proper Class AB waveform output from the LM386N stage. Hardware Prototyping: The complete circuit was assembled on a breadboard and powered from a regulated DC supply, and tested for FM reception and audio output using a 75 cm wire antenna. Finally, it was soldered on the solid board and tested as well.		
Achievements and Skills Gained	• Understood FM signal reception, LC resonance-based tuning, and FM demodulation using the TDA7000 IC. • Designed the LC tank circuit from first principles using $f_0 = 1/(2\pi\sqrt{LC})$, calculating the required inductance for the FM band with the selected variable capacitor. • Applied LTspice simulation to verify circuit behavior before hardware construction. • Designed a completely regulated AC-to-DC power supply as a bonus subsystem. • Learned solid board layout principles and soldering techniques for hardware integration.		
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1.Component list with specification

Component	Value / Model	Role in Circuit	Qty
TDA7000 IC	18-pin DIP	FM reception: RF amp, mixer, IF filter, demodulator	1
LM386N IC	8-pin DIP	Class AB audio amplifier — push-pull output stage	1
Inductor L1	~40.5 nH	LC tank oscillator for FM station tuning	1
Variable Cap C1	0–60 pF	FM station selection across 88–108 MHz	1
Ceramic Cap 221	220 pF	RF bypass / filter on TDA7000 pins	2
Ceramic Cap 331	330 pF	RF bypass / filter on TDA7000 pins	2
Ceramic Cap 181	180 pF	RF bypass / filter on TDA7000 pins	2
Ceramic Cap 151	150 pF	RF bypass / filter on TDA7000 pins	1
Ceramic Cap 104	0.1 μF	Decoupling / supply bypass	4
Ceramic Cap 103	0.01 μF	Decoupling	1
Ceramic Cap 102	0.001 μF	Decoupling	1
Ceramic Cap 223	0.022 μF	Decoupling	1
Ceramic Cap 332	0.0033 μF	Decoupling	1
Electrolytic Cap	220 μF	Power supply filtering (×2)	2
Electrolytic Cap	4.7 μF	Supply decoupling at LM386N	1
Resistor	22 kΩ	TDA7000 pin 2 bias	1
Potentiometer	10 kΩ	Volume control (LM386N pin 3)	1
Resistor	10 Ω	LM386N output stability	1
Speaker	8 Ω	Audio output transducer	1
Wire Antenna	~75 cm	Quarter-wave at 100 MHz	1
Transformer	230V → 12V AC	Steps down mains voltage	1
Bridge Rectifier	4× 1N4148	Full-wave rectification	1
Electrolytic Cap	2200 μF	Smoothing filter after rectifier	1
Ceramic Cap	0.1 μF ×2	Input/output bypass for 7809	2
7809 Regulator	+9V, TO-220	Regulated 9V DC output	1
Zener Diode	9V	Output overvoltage protection	1
Resistor	560 Ω	Zener current-limiting	1
Diode	1N4148	Reverse polarity protection	1

Main Results

2. Complete System Schematic

Signal flow: 75 cm antenna → TDA7000 FM Receiver IC (LC tank tunes station, internal demodulator extracts audio) → volume control potentiometer → LM386N Class AB amplifier → 8Ω speaker. Powered by the regulated 9V DC supply.

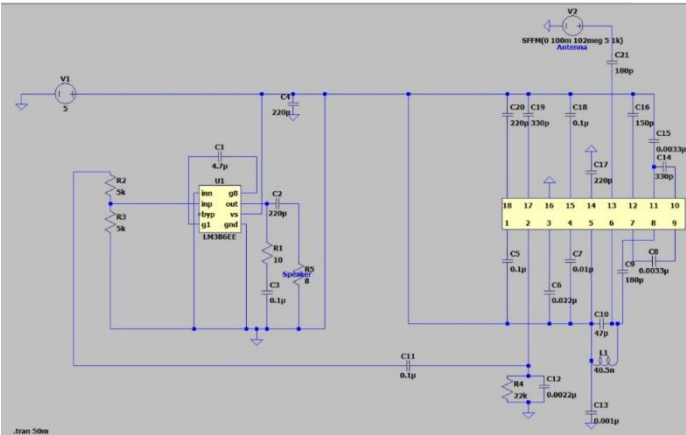


Fig. 1 — Complete FM Radio Receiver Schematic (Antenna → TDA7000 → LM386N → Speaker)

3. Class AB Amplifier — LM386N

The LM386N implements Class AB amplification internally. Its output stage uses complementary NPN/PNP transistors in a push-pull configuration biased with a small quiescent current, preventing both transistors from switching fully off at the audio zero-crossing. This eliminates crossover distortion. Default gain is 20. The 10Ω output resistor ensures stability; the 220μF coupling capacitor blocks DC from the speaker. The final gain is 2/0.1=20

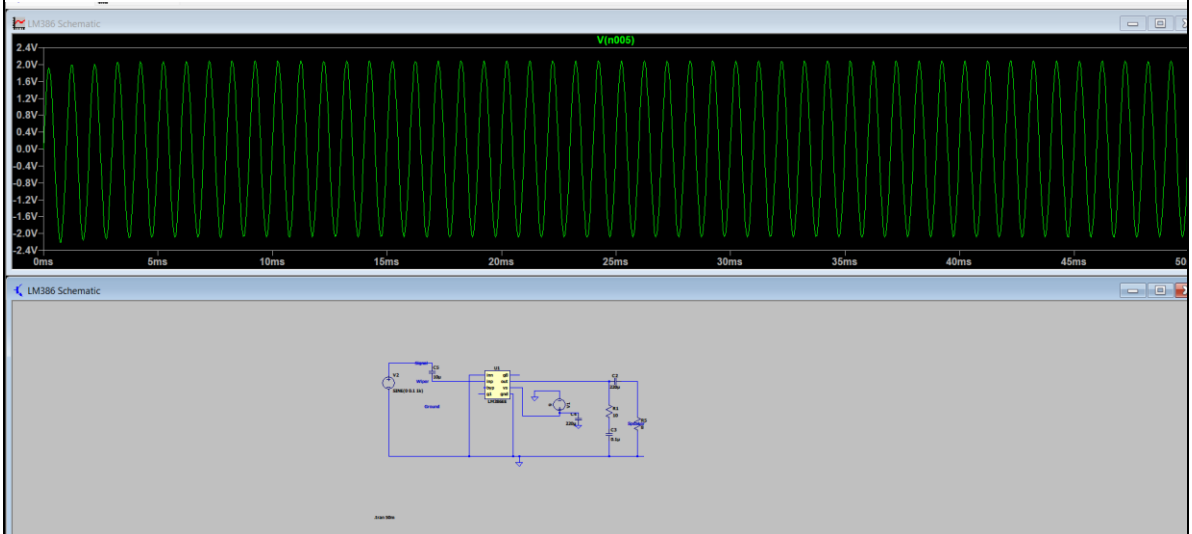


Fig. 2 — LM386N Class AB Amplifier: LTspice Simulation (Input vs. Output)

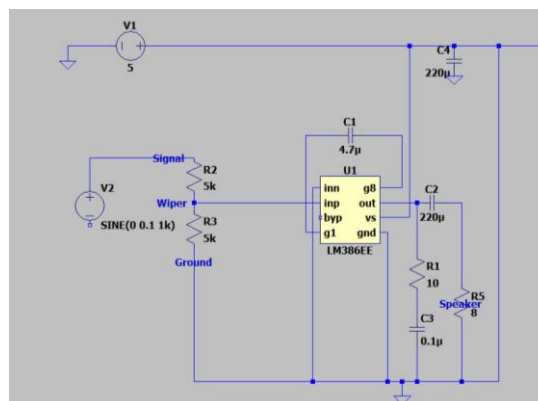


Fig. 3 — LM386N Class AB Amplifier Circuit Schematic (LTspice)

4. LC Tuning Circuit

$$f_0 = 1/(2\pi\sqrt{LC}) \rightarrow \text{from } L = 40.5 \text{ nH } C = 10\text{-}60 \text{ pF (for calculating)}$$

$$f_1 = 102 \text{ Mhz, } f_2 = 250 \text{ Mhz}$$

$$Q = \omega L/R = 102 \text{ (at } f=100\text{Mhz and } R=0.25\Omega)$$

This is realized by winding 3.5 turns of enameled copper wire on a 6 mm former. Tuning range: ~92 MHz (C=60 pF) to 108 MHz (C=50 pF), covering the full FM broadcast band. The 75 cm antenna is a quarter-wavelength at 100 MHz ($\lambda/4 = c/4f = 0.75 \text{ m}$), maximizing signal coupling.

5. Bonus: Regulated AC Power Supply

A regulated AC-to-DC power supply was designed as a bonus subsystem, replacing the 9V battery with a stable mains-powered source:

1. **Transformer:** Steps down 230V mains AC to 12V AC, providing isolation and safe working voltage.
2. **Bridge Rectifier (4× 1N4007):** Full-wave rectification converts AC to pulsating DC
3. **Filter (2200 μF + 0.1 μF):** Smooths pulsating DC.
4. **7809 Regulator:** Regulates to precise +9V DC regardless of load or input fluctuations. Input/output 0.1 μF bypass capacitors ensure stability.
5. **Protection:** 9V zener + 560Ω clamp overvoltage. 1N4148 diode prevents reverse polarity. Fuse protects mains input.

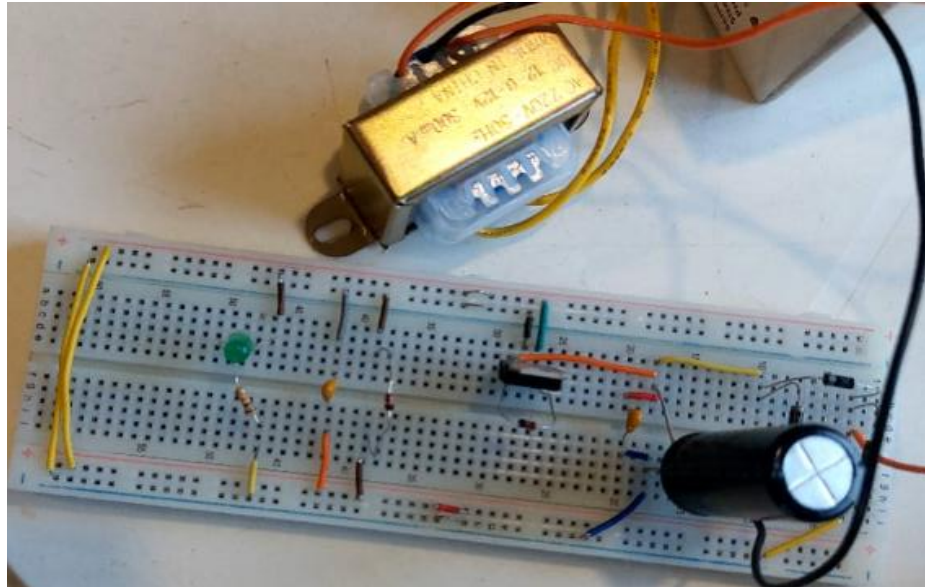
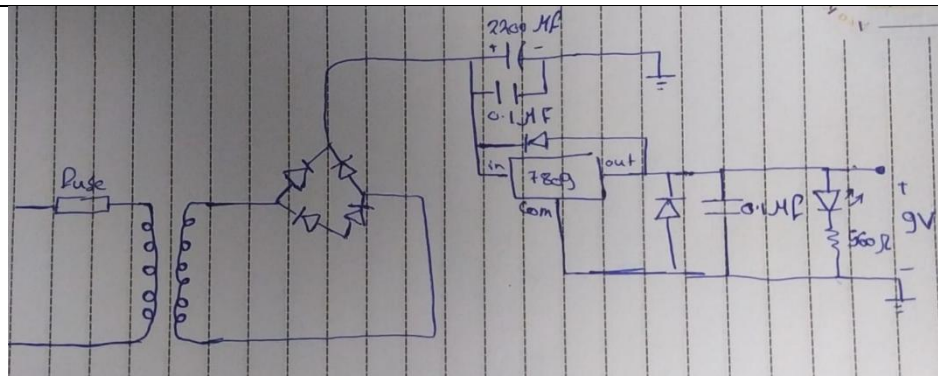


Fig. 6 — Bonus: Regulated AC-to-DC Power Supply Schematic and implementation (Transformer → Rectifier → 7809 → 9V)

6. Hardware Implementation

The complete circuit was assembled on a breadboard. The TDA7000 and LM386N ICs were placed with all external passive components per the schematic. A 75 cm wire antenna was connected to pin 13. The circuit was powered from the regulated DC supply (9V). FM broadcast signals were successfully received.

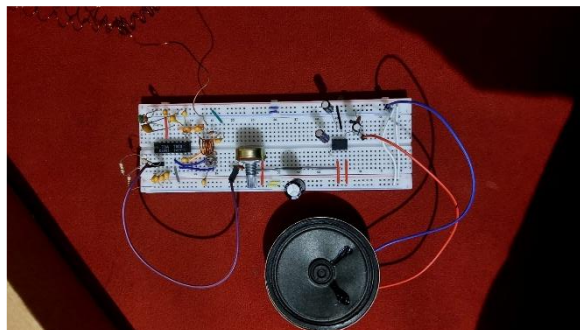


Fig. 7 — Breadboard Implementation: Complete FM Radio Receiver System

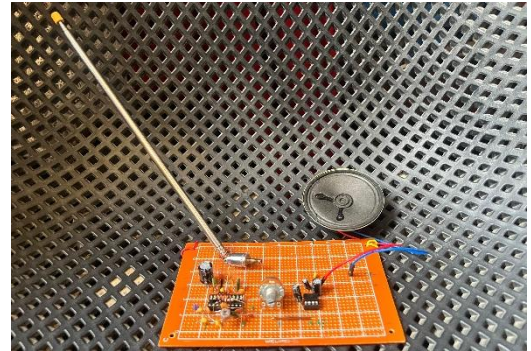
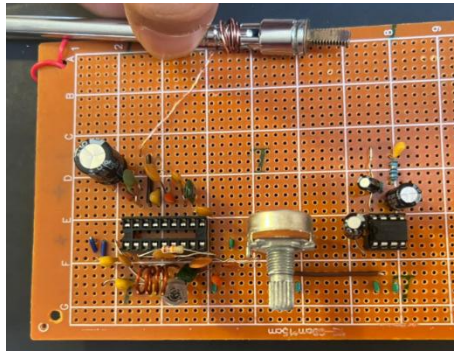


Fig. 7 — Solidboard Implementation: Complete FM Radio Receiver System

1. FM Tunable Receiver — TDA7000 and LC Tank

The TDA7000 provides a complete FM receiver front-end in a single IC. Station selection is controlled by the external LC tank: varying C_1 shifts the VCO frequency. Smaller $C \rightarrow$ higher frequency (higher station); larger $C \rightarrow$ lower frequency. The 3.5-turn coil (~ 40.5 nH) with 0-60 pF total capacitance centers the tuning range correctly across the FM band. Component placement is critical at FM frequencies — the coil must be kept away from metal and shielded if possible to prevent inductance drift and frequency instability.

2. Class AB Audio Amplification — LM386N

The LM386N satisfies Class AB requirement through its internal architecture. The output NPN/PNP pair is biased into slight simultaneous conduction, preventing full switch-off at the audio zero-crossing, eliminating crossover distortion. The volume potentiometer on pin 3 controls the audio level before amplification. The 10 Ω series output resistor suppresses oscillation from capacitive loading, and the 220 μ F coupling capacitor protects the speaker from DC bias.

Comparison Table between output in Simulation and Oscilloscope

Point of Comparison	Simulation	Oscilloscope
Input	0.1v	10mv
output	2v	296mv
Quality factor	20	29.6

The difference between the simulation and oscilloscope results for the LM386 Audio Amplifier IC is in the normal range. In simulation, the circuit is ideal and does not include real-world effects such as wiring resistance, noise, or component tolerances, so the gain usually stays close to the theoretical value of about 20. However, in the oscilloscope, the circuit behaves in a real environment where parasitic effects from connections, small variations in component values, and measurement inaccuracies can affect the results. These practical factors can cause the measured gain to increase or slightly deviate from the expected value. Therefore, obtaining a gain of 29.6 in the breadboard compared to 20 in simulation is reasonable and expected in real experimental setups.

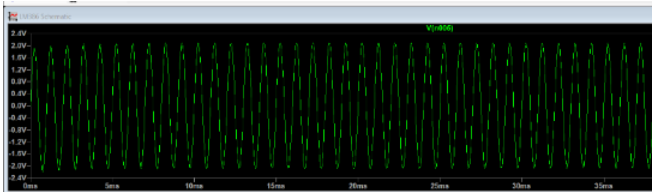


Fig.8 the output on LTspice

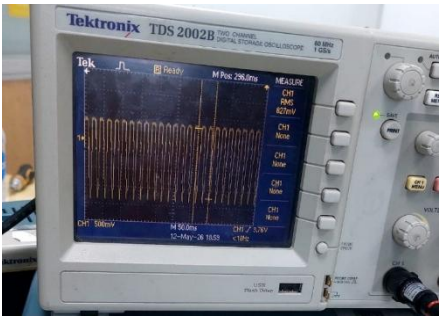


Fig.9 the output on Oscilloscope

Discussion
and
Conclusion

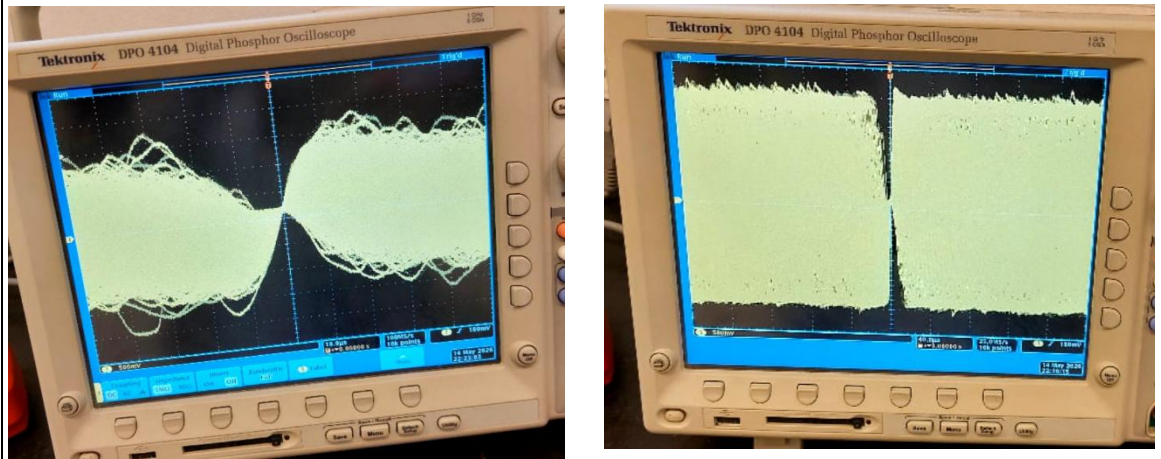


Fig.10 the output on the oscilloscope (The amplitude varies due to rotating the potentiometer)

Conclusion

The FM radio receiver system successfully demonstrates a complete three-stage signal chain: FM signal capture via a 75 cm quarter-wave antenna, frequency-selective demodulation via the TDA7000 and LC tank, and Class AB audio amplification via the LM386N driving an 8Ω speaker. As a bonus, a fully regulated AC-to-DC power supply was designed and added to the system. The LC tank design grounded in $f_0 = 1/(2\pi\sqrt{LC})$ provides the selectivity needed for clear FM reception. The LM386N's internal push-pull Class AB output stage delivers efficient, low-distortion audio. Hardware implementation on a breadboard confirmed the viability of the complete design

References

1. Alexander, C. K., & Sadiku, M. N. O. (2012). Fundamentals of Electric Circuits (5th ed.). McGraw-Hill.
2. Kennedy, G., & Davis, B. (1992). Electronic Communication Systems. McGraw-Hill.
3. Sedra, A., & Smith, K. (2019). Microelectronic Circuits (8th ed.). Oxford University Press.

Future Work and Suggestions

- Replace the manual variable capacitor with a varactor diode for electronic/digital station tuning and AFC (automatic frequency control).
- Implement an RSSI signal-strength indicator using the TDA7000 mute output to help with antenna positioning.
- Add heatsinking to the 7809 voltage regulator and investigate switching-regulator alternatives to reduce heat dissipation at higher load currents.